

Stokes Layer Potentials to Laplace Layer Potential Derivations

Brian Ha

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1 Laplace Layer Potentials

The Laplace scalar layer potentials are given as follows,

$$\begin{aligned} S^L[\sigma](\mathbf{x}) &= \int_{\Gamma} G^L(\mathbf{x}, \mathbf{y}) \sigma(\mathbf{y}) d\Gamma(\mathbf{y}) = \int_{\Gamma} \frac{1}{4\pi|\mathbf{x} - \mathbf{y}|} \sigma(\mathbf{y}) d\Gamma(\mathbf{y}) \\ D^L[\mu](\mathbf{x}) &= \int_{\Gamma} \frac{\partial}{\partial \mathbf{n}(\mathbf{y})} G^L(\mathbf{x}, \mathbf{y}) \mu(\mathbf{y}) d\Gamma(\mathbf{y}) = \int_{\Gamma} \frac{(\mathbf{x} - \mathbf{y}) \cdot \mathbf{n}(\mathbf{y})}{4\pi|\mathbf{x} - \mathbf{y}|^3} \mu(\mathbf{y}) d\Gamma(\mathbf{y}) \end{aligned}$$

2 Stokes Double-Layer Potential

As a remark, the identity used in Bagge's thesis [1] uses a different definition of the Laplace scalar double layer potential:

$$D_{\text{alt}}^L[\vec{\phi}](\mathbf{x}) = \int_{\Gamma} \vec{\phi}(\mathbf{y}) \cdot \nabla G(\mathbf{x}, \mathbf{y}) dS(\mathbf{y}) = \int_{\Gamma} \vec{\phi}(\mathbf{y}) \cdot \nabla \left(\frac{1}{4\pi\|\mathbf{r}\|} \right) dS(\mathbf{y})$$

However, this does not work with the code, as is, and also would probably lead to complications in the implementation since the jump relations are no longer obvious.

Even if one did try to work with this definition and tried to reformulate the definition to the standard one, higher-order derivatives would start to appear. This issue is also seems to be present if one tries to work with the identities given in Tornberg et. al [5].

Instead, we consider an alternative derivation that uses the standard definition of the Laplace DLP. Recall the Stokes double-layer potential (DLP),

$$\mathcal{D}[\vec{\sigma}](\mathbf{x}) = \frac{3}{4\pi} \int_{\Gamma} \left(\frac{\mathbf{r} \cdot \mathbf{n}(\mathbf{y})}{\|\mathbf{r}\|^2} \mathbf{r} \right) \left(\frac{\mathbf{r} \cdot \sigma(\mathbf{y})}{\|\mathbf{r}\|^3} \right) d\Gamma(\mathbf{y})$$

The idea is to use the following identity, which is inspired by Barnett et. al [2],

$$\begin{aligned} \frac{\mathbf{r} \cdot \mathbf{n}(\mathbf{y})}{\|\mathbf{r}\|^2} \mathbf{r} &= \frac{\|\mathbf{r}\|^3}{3} \left[\frac{\mathbf{n}(\mathbf{y})}{\|\mathbf{r}\|^3} - \nabla_x \left(\frac{\mathbf{r} \cdot \mathbf{n}(\mathbf{y})}{\|\mathbf{r}\|^3} \right) \right] \\ &= \frac{1}{3} \left[\mathbf{n}(\mathbf{y}) - \|\mathbf{r}\|^3 \nabla_x \left(\frac{\mathbf{r} \cdot \mathbf{n}(\mathbf{y})}{\|\mathbf{r}\|^3} \right) \right] \end{aligned}$$

The above equation is derived from

$$\nabla_x \left(\frac{\mathbf{r} \cdot \mathbf{n}(\mathbf{y})}{\|\mathbf{r}\|^3} \right) = \frac{\mathbf{n}(\mathbf{y})}{\|\mathbf{r}\|^3} - 3 \frac{\mathbf{r} \cdot \mathbf{n}(\mathbf{y})}{\|\mathbf{r}\|^5} \mathbf{r}.$$

Re-ordering the above equation yields

$$\begin{aligned}\frac{r \cdot n(\mathbf{y})}{\|r\|^2} r &= \frac{\|r\|^3}{3} \left[\frac{n(\mathbf{y})}{\|r\|^3} - \nabla_x \left(\frac{r \cdot n(\mathbf{y})}{\|r\|^3} \right) \right] \\ &= \frac{1}{3} \left[n(\mathbf{y}) - \|r\|^3 \nabla_x \left(\frac{r \cdot n(\mathbf{y})}{\|r\|^3} \right) \right]\end{aligned}$$

Plugging this expression into our layer potential, we obtain

$$\begin{aligned}\mathcal{D}[\sigma](\mathbf{x}) &= \frac{3}{4\pi} \int_{\Gamma} \left(\frac{r \cdot n(\mathbf{y})}{\|r\|^2} r \right) \left(\frac{r \cdot \sigma(\mathbf{y})}{\|r\|^3} \right) d\Gamma(\mathbf{y}) \\ &= \frac{1}{4\pi} \int_{\Gamma} \left[n(\mathbf{y}) - \|r\|^3 \nabla_x \left(\frac{r \cdot n(\mathbf{y})}{\|r\|^3} \right) \right] \frac{r \cdot \sigma(\mathbf{y})}{\|r\|^3} d\Gamma(\mathbf{y}) \\ &= -\frac{1}{4\pi} \left[\int_{\Gamma} \nabla_x \left(\frac{r \cdot n(\mathbf{y})}{\|r\|^3} \right) (r \cdot \sigma(\mathbf{y})) d\Gamma(\mathbf{y}) - \int_{\Gamma} \frac{n(\mathbf{y})}{\|r\|^3} (r \cdot \sigma(\mathbf{y})) d\Gamma(\mathbf{y}) \right]\end{aligned}$$

Let's handle each term separately. For the first term, observe that

$$\begin{aligned}\int_{\Gamma} \nabla_x \left(\frac{r \cdot n(\mathbf{y})}{\|r\|^3} \right) (r \cdot \sigma(\mathbf{y})) d\Gamma(\mathbf{y}) &= \int_{\Gamma} \nabla_x \left(\frac{r \cdot n(\mathbf{y})}{\|r\|^3} \right) ((\mathbf{x} - \mathbf{y}) \cdot \sigma(\mathbf{y})) d\Gamma(\mathbf{y}) \\ &= \int_{\Gamma} \nabla_x \left(\frac{r \cdot n(\mathbf{y})}{\|r\|^3} \right) (\mathbf{x} \cdot \sigma(\mathbf{y})) d\Gamma(\mathbf{y}) \\ &\quad - \int_{\Gamma} \nabla_x \left(\frac{r \cdot n(\mathbf{y})}{\|r\|^3} \right) (\mathbf{y} \cdot \sigma(\mathbf{y})) d\Gamma(\mathbf{y}) \\ &= \sum_{k=1}^3 x_k \int_{\Gamma} \nabla_x \left(\frac{r \cdot n(\mathbf{y})}{\|r\|^3} \right) \sigma_k(\mathbf{y}) d\Gamma(\mathbf{y}) - 4\pi \nabla_x D^L[\mathbf{y} \cdot \sigma](\mathbf{x}) \\ &= 4\pi \left(\sum_{k=1}^3 x_k \nabla_x D^L[\sigma_k](\mathbf{x}) \right) - 4\pi \nabla_x D^L[\mathbf{y} \cdot \sigma](\mathbf{x})\end{aligned}$$

Next, we handle the second term. Define

$$I(x) = \int_{\Gamma} \frac{n(\mathbf{y})}{\|r\|^3} (r \cdot \sigma(\mathbf{y})) d\Gamma(\mathbf{y})$$

Now, let's examine the k -th component of this vector I , which gives us

$$\begin{aligned}
I_k(x) &= \int_{\Gamma} \frac{n_k(\mathbf{y})}{\|r\|^3} (r \cdot \sigma(\mathbf{y})) dx \\
&= - \int_{\Gamma} n_k(\mathbf{y}) \left(\nabla_x \left(\frac{1}{\|r\|} \right) \cdot \sigma(\mathbf{y}) \right) d\Gamma(\mathbf{y}) \\
&= - \sum_{j=1}^3 \int_{\Gamma} n_k(\mathbf{y}) \left(\frac{\partial}{\partial x_j} \frac{1}{\|r\|} \right) \sigma_j(\mathbf{y}) d\Gamma(\mathbf{y}) \\
&= - \sum_{j=1}^3 \frac{\partial}{\partial x_j} \int_{\Gamma} \frac{n_k(\mathbf{y}) \sigma_j(\mathbf{y})}{\|r\|} d\Gamma(\mathbf{y}) \\
&= -4\pi \sum_{j=1}^3 e_j \cdot \nabla S^L[n_k(\cdot) \sigma_j(\cdot)](\mathbf{x})
\end{aligned}$$

Thus, summarizing our work, we have the following representation for the Stokes double layer potential,

$$\mathcal{D}[\sigma](\mathbf{x}) = - \left(\sum_{k=1}^3 x_k \nabla_x D^L[\sigma_k(\cdot)](\mathbf{x}) \right) + \nabla_x D^L[\mathbf{y} \cdot \sigma(\cdot)](\mathbf{x}) + I(\mathbf{x})$$

which we may re-write as

$$\begin{aligned}
\mathcal{D}[\sigma](\mathbf{x})_i &= \mathbf{e}_i \cdot \nabla_x D^L[\mathbf{y} \cdot \sigma(\cdot)](\mathbf{x}) - \left(\sum_{k=1}^3 x_k \mathbf{e}_i \cdot \nabla_x D^L[\sigma_k(\cdot)](\mathbf{x}) \right) \\
&\quad - \sum_{j=1}^3 \mathbf{e}_j \cdot \nabla S^L[n_i(\cdot) \sigma_j(\cdot)](\mathbf{x}).
\end{aligned}$$

3 Traction of Single-Layer Potential

Recall the traction of the Stokes single-layer potential (TSL):

$$\mathcal{K}[\vec{\sigma}](\mathbf{x}) = -\frac{3}{4\pi} \int_{\Gamma} \left(\frac{r \cdot \vec{\sigma}(\mathbf{y})}{\|r\|^2} r \right) \left(\frac{r \cdot n(\mathbf{x})}{\|r\|^3} \right) d\Gamma(\mathbf{y})$$

Now, observe that

$$\nabla_x \left(\frac{r \cdot \sigma(\mathbf{y})}{\|r\|^3} \right) = \frac{\sigma(\mathbf{y})}{\|r\|^3} - 3 \frac{r \cdot \sigma(\mathbf{y})}{\|r\|^5} r$$

Re-ordering the above equation yields

$$\begin{aligned}
\frac{r \cdot \sigma(\mathbf{y})}{\|r\|^2} r &= \frac{\|r\|^3}{3} \left[\frac{\sigma(\mathbf{y})}{\|r\|^3} - \nabla_x \left(\frac{r \cdot \sigma(\mathbf{y})}{\|r\|^3} \right) \right] \\
&= \frac{1}{3} \left[\sigma(\mathbf{y}) - \|r\|^3 \nabla_x \left(\frac{r \cdot \sigma(\mathbf{y})}{\|r\|^3} \right) \right]
\end{aligned}$$

Plugging this into the traction layer potential,

$$\begin{aligned}\mathcal{K}[\sigma](\mathbf{x}) &= -\frac{3}{4\pi} \int_{\Gamma} \left(\frac{r \cdot \sigma(\mathbf{y})}{\|r\|^2} r \right) \left(\frac{r \cdot n(\mathbf{x})}{\|r\|^3} \right) d\Gamma(\mathbf{y}) \\ &= \frac{1}{4\pi} \int_{\Gamma} \nabla_x \left(\frac{r \cdot \sigma(\mathbf{y})}{\|r\|^3} \right) (r \cdot n(\mathbf{x})) - \frac{\sigma(\mathbf{y})}{\|r\|^3} (r \cdot n(\mathbf{x})) d\Gamma(\mathbf{y})\end{aligned}$$

As before, there are two terms to consider. Let's start with the first term, where use that $\mathbf{r} = \mathbf{x} - \mathbf{y}$,

$$\begin{aligned}\int_{\Gamma} \nabla_x \left(\frac{r \cdot \sigma(\mathbf{y})}{\|r\|^3} \right) (r \cdot n(\mathbf{x})) d\Gamma(\mathbf{y}) &= (\mathbf{x} \cdot n(\mathbf{x})) \int_{\Gamma} \nabla_x \left(\frac{r \cdot \sigma(\mathbf{y})}{\|r\|^3} \right) d\Gamma(\mathbf{y}) \\ &\quad - \int_{\Gamma} \nabla_x \left(\frac{r \cdot \sigma(\mathbf{y})}{\|r\|^3} \right) (\mathbf{y} \cdot n(\mathbf{x})) d\Gamma(\mathbf{y})\end{aligned}$$

For the x -term, we have

$$\begin{aligned}(\mathbf{x} \cdot n(\mathbf{x})) \int_{\Gamma} \nabla_x \left(\frac{r \cdot \sigma(\mathbf{y})}{\|r\|^3} \right) d\Gamma(\mathbf{y}) &= (\mathbf{x} \cdot n(\mathbf{x})) \nabla_x \int_{\Gamma} \frac{r \cdot \sigma(\mathbf{y})}{\|r\|^3} d\Gamma(\mathbf{y}) \\ &= -(\mathbf{x} \cdot n(\mathbf{x})) \nabla_x \int_{\Gamma} \nabla_x \left(\frac{1}{\|r\|} \right) \cdot \sigma(\mathbf{y}) d\Gamma(\mathbf{y}) \\ &= -(\mathbf{x} \cdot n(\mathbf{x})) \nabla_x \left[\sum_{k=1}^3 \frac{\partial}{\partial x_k} \int_{\Gamma} \frac{\sigma_k(\mathbf{y})}{\|r\|} d\Gamma(\mathbf{y}) \right] \\ &= -4\pi(\mathbf{x} \cdot n(\mathbf{x})) \nabla_x \left[\sum_{k=1}^3 e_k \cdot \nabla_x S^L[\sigma_k](\mathbf{x}) \right] \\ &= -4\pi(\mathbf{x} \cdot n(\mathbf{x})) \nabla_x [\operatorname{div}_x S^L[\sigma](\mathbf{x})]\end{aligned}$$

For the y -term,

$$\begin{aligned}\int_{\Gamma} \nabla_x \left(\frac{r \cdot \sigma(\mathbf{y})}{\|r\|^3} \right) (\mathbf{y} \cdot n(\mathbf{x})) d\Gamma(\mathbf{y}) &= \sum_{j=1}^3 n_j(\mathbf{x}) \int_{\Gamma} \nabla_x \left(\frac{r \cdot \sigma(\mathbf{y})}{\|r\|^3} \right) y_j d\Gamma(\mathbf{y}) \\ &= \sum_{j=1}^3 n_j(\mathbf{x}) \nabla_x \int_{\Gamma} \left(\frac{r \cdot \sigma(\mathbf{y})}{\|r\|^3} \right) y_j d\Gamma(\mathbf{y}) \\ &= -\sum_{j=1}^3 n_j(\mathbf{x}) \nabla_x \int_{\Gamma} \left[\nabla_x \left(\frac{1}{\|r\|} \right) \cdot \sigma(\mathbf{y}) \right] y_j d\Gamma(\mathbf{y}) \\ &= -\sum_{j=1}^3 n_j(\mathbf{x}) \nabla_x \left[\sum_{k=1}^3 \frac{\partial}{\partial x_k} \int_{\Gamma} \frac{\sigma_k(\mathbf{y}) y_j}{\|r\|} d\Gamma(\mathbf{y}) \right] \\ &= -4\pi \sum_{j=1}^3 n_j(\mathbf{x}) \nabla_x \left[\sum_{k=1}^3 e_k \cdot \nabla S^L[y_j \sigma_k(\cdot)](\mathbf{x}) \right] \\ &= -4\pi \sum_{j=1}^3 n_j(x) \nabla_x (\operatorname{div}_x S^L[y_j \sigma](\mathbf{x}))\end{aligned}$$

Thus,

$$\int_{\Gamma} \nabla_x \left(\frac{r \cdot \sigma(\mathbf{y})}{\|r\|^3} \right) (r \cdot n(\mathbf{x})) d\Gamma(\mathbf{y}) = 4\pi \left[\sum_{j=1}^3 n_j(x) \nabla_x (\operatorname{div}_x S^L[y_j \sigma](\mathbf{x})) \right] - 4\pi (\mathbf{x} \cdot n(\mathbf{x})) \nabla_x [\operatorname{div}_x S^L[\sigma](\mathbf{x})]$$

Next, we consider the second term from above:

$$\begin{aligned} \int_{\Gamma} \frac{\sigma(\mathbf{y})}{\|r\|^3} (r \cdot n(\mathbf{x})) d\Gamma(\mathbf{y}) &= - \int_{\Gamma} \left(\nabla_x \left(\frac{1}{\|r\|} \right) \cdot n(x) \right) \sigma(\mathbf{y}) d\Gamma(\mathbf{y}) \\ &= - \sum_{k=1}^3 n_k(\mathbf{x}) \frac{\partial}{\partial x_k} \int_{\Gamma} \frac{1}{\|r\|} \sigma(\mathbf{y}) d\Gamma(\mathbf{y}) \\ &= -4\pi \sum_{k=1}^3 n_k(\mathbf{x}) \left(\frac{\partial}{\partial x_k} S^L[\sigma](\mathbf{x}) \right) \end{aligned}$$

Thus, we yield the following formula that utilizes second-order derivatives (and note the distinction between the scalar and vector layer potentials based on the densities),

$$\begin{aligned} \mathcal{K}[\vec{\sigma}](\mathbf{x})_i &= \sum_{k=1}^3 n_k(\mathbf{x}) (\mathbf{e}_k \cdot \nabla S^L[\sigma_i](\mathbf{x})) + \sum_{j=1}^3 n_j(x) [\mathbf{e}_i \cdot \nabla_x (\operatorname{div}_x S^L[y_j \vec{\sigma}](\mathbf{x}))] \\ &\quad - (\mathbf{x} \cdot n(\mathbf{x})) (\mathbf{e}_i \cdot \nabla_x [\operatorname{div}_x S^L[\vec{\sigma}](\mathbf{x})]) \end{aligned}$$

3.1 Derivation of First-Order Formula

To remove the use of these higher-order derivatives, we utilize the following formula (over a smooth surface—i.e. a spheroid); a reference for this is found in Colton et al. [3].

$$\nabla_x \left(\nabla_x \cdot \int_{\Gamma} G(\mathbf{x}, \mathbf{y}) \vec{\sigma}(\mathbf{y}) dS(\mathbf{y}) \right) = \nabla_x S^L[\nabla_{\Gamma} \cdot \vec{\sigma}](\mathbf{x}) - \nabla_x D^L[n \cdot \vec{\sigma}](\mathbf{x})$$

where we recall that

$$\operatorname{div}_x S^L[\vec{\sigma}](\mathbf{x}) = \nabla_x \cdot \int_{\Gamma} G(\mathbf{x}, \mathbf{y}) \vec{\sigma}(\mathbf{y}) dS(\mathbf{y})$$

To derive the formula above, we note that we may re-write the divergence of the Laplace vector SLP as follows,

$$\nabla_x \cdot \int_{\Gamma} G(\mathbf{x}, \mathbf{y}) \vec{\sigma}(\mathbf{y}) dS(\mathbf{y}) = \int_{\Gamma} (\nabla_x G(\mathbf{x}, \mathbf{y})) \cdot \vec{\sigma}(\mathbf{y}) dS(\mathbf{y})$$

where we used $\operatorname{div}(U \mathbf{A}) = U(\operatorname{div} \mathbf{A}) + \mathbf{A} \cdot \operatorname{grad} U$.

Now, we use the following property,

$$\nabla_x G(\mathbf{x}, \mathbf{y}) = -\nabla_y G(\mathbf{x}, \mathbf{y})$$

to obtain

$$\nabla_x \cdot \int_{\Gamma} G(\mathbf{x}, \mathbf{y}) \vec{\sigma}(\mathbf{y}) dS(\mathbf{y}) = - \int_{\Gamma} (\nabla_y G(\mathbf{x}, \mathbf{y})) \cdot \vec{\sigma}(\mathbf{y}) dS(\mathbf{y})$$

Now, consider the decomposition of both the vectors $\nabla_y G(\mathbf{x}, \mathbf{y})$ and $\vec{\sigma}(\mathbf{y})$ at the point $\mathbf{y}$, where we may split the both of them into their normal and tangetial components on the surface:

$$\begin{aligned} \vec{\sigma}(\mathbf{y}) &= \vec{\sigma}(\mathbf{y})_{\perp} + (n(\mathbf{y}) \cdot \vec{\sigma}(\mathbf{y}))n(\mathbf{y}) \\ \nabla_y G(\mathbf{x}, \mathbf{y}) &= \nabla_{\Gamma,y} G(\mathbf{x}, \mathbf{y}) + \frac{\partial G(\mathbf{x}, \mathbf{y})}{\partial n(\mathbf{y})} n(\mathbf{y}) \end{aligned}$$

where $\nabla_{\Gamma,y}$ denotes the surface gradient operator and $\vec{\sigma}(\mathbf{y})_{\perp}$ is the perpendicular component of the vector density on the surface.

Here, we assume that the vector $n(\mathbf{y})$ is a unit normal vector: using this yields

$$(\nabla_y G(\mathbf{x}, \mathbf{y})) \cdot \vec{\sigma}(\mathbf{y}) = \vec{\sigma}(\mathbf{y})_{\perp} \cdot \nabla_{\Gamma,y} G(\mathbf{x}, \mathbf{y}) + \frac{\partial G(\mathbf{x}, \mathbf{y})}{\partial n(\mathbf{y})} (n(\mathbf{y}) \cdot \vec{\sigma}(\mathbf{y}))$$

Hence,

$$\nabla_x \cdot \int_{\Gamma} G(\mathbf{x}, \mathbf{y}) \vec{\sigma}(\mathbf{y}) dS(\mathbf{y}) = - \int_{\Gamma} \vec{\sigma}(\mathbf{y})_{\perp} \cdot \nabla_{\Gamma,y} G(\mathbf{x}, \mathbf{y}) dS(\mathbf{y}) - \int_{\Gamma} \frac{\partial G(\mathbf{x}, \mathbf{y})}{\partial n(\mathbf{y})} (n(\mathbf{y}) \cdot \vec{\sigma}(\mathbf{y})) dS(\mathbf{y})$$

Note that the RHS can be broken up into two separate terms. Let's handle the second term as that's more obvious: it's just the Laplace (scalar) double layer potential $D^L[n \cdot \sigma](\mathbf{x})$. Now, let's handle the first term:

$$\int_{\Gamma} \vec{\sigma}(\mathbf{y})_{\perp} \cdot \nabla_{\Gamma,y} G(\mathbf{x}, \mathbf{y}) dS(\mathbf{y})$$

Recall the divergence product rule

$$\nabla_{\Gamma,y} \cdot (G(\mathbf{x}, \mathbf{y}) \vec{\sigma}(\mathbf{y})_{\perp}) = G(\mathbf{x}, \mathbf{y}) (\nabla_{\Gamma} \cdot \vec{\sigma}(\mathbf{y})_{\perp}) + \vec{\sigma}(\mathbf{y})_{\perp} \cdot (\nabla_{\Gamma,y} G(\mathbf{x}, \mathbf{y}))$$

From the divergence theorem, for any vector field $\mathbf{F}$ on a closed, smooth surface Γ , we have

$$\int_{\Gamma} \nabla_{\Gamma,y} \cdot (G(\mathbf{x}, \mathbf{y}) \vec{\sigma}(\mathbf{y})_{\perp}) dS = 0$$

Thus, plugging the product rule into the above, we yield

$$\begin{aligned} \int_{\Gamma} \vec{\sigma}(\mathbf{y})_{\perp} \cdot (\nabla_{\Gamma,y} G(\mathbf{x}, \mathbf{y})) dS(\mathbf{y}) &= - \int_{\Gamma} G(\mathbf{x}, \mathbf{y}) (\nabla_{\Gamma,y} \cdot \vec{\sigma}(\mathbf{y})_{\perp}) dS(\mathbf{y}) \\ &= -S^L[\nabla_{\Gamma,y} \cdot \vec{\sigma}(\mathbf{y})_{\perp}](\mathbf{x}) \end{aligned}$$

where we used that $\nabla_{\Gamma,y} \cdot \vec{\sigma}(\mathbf{y})_{\perp} = \nabla_{\Gamma,y} \cdot \vec{\sigma}(\mathbf{y})$ for an appropriate definition of the surface divergence. We shall use the following definition (a reference for this may be found in Grinfeld [4]): for a curvilinear coordinate system (u, v, ϕ) , over a surface $u = u_0$, the surface divergence is given by

$$\nabla_S \cdot \mathbf{F} = \frac{1}{h_v h_\phi} \left[\frac{\partial}{\partial v} (h_\phi F_v) + \frac{\partial}{\partial \phi} (h_v F_\phi) \right]$$

where F_v and F_ϕ represents the components of the vector F . From this, we conclude that

$$\begin{aligned} \nabla_x \cdot \int_{\Gamma} G(\mathbf{x}, \mathbf{y}) \vec{\sigma}(\mathbf{y}) dS(\mathbf{y}) &= - \underbrace{\int_{\Gamma} \vec{\sigma}(\mathbf{y})_{\perp} \cdot \nabla_{\Gamma,y} G(\mathbf{x}, \mathbf{y}) dS(\mathbf{y})}_{-S^L[\nabla_{\Gamma,y} \cdot \vec{\sigma}(\mathbf{y})](\mathbf{x})} \\ &\quad - \underbrace{\int_{\Gamma} \frac{\partial G(\mathbf{x}, \mathbf{y})}{\partial n(\mathbf{y})} (n(\mathbf{y}) \cdot \vec{\sigma}(\mathbf{y})) dS(\mathbf{y})}_{D^L[n \cdot \vec{\sigma}](\mathbf{x})} \end{aligned}$$

or re-writing the above expression,

$$\operatorname{div}_x S^L[\vec{\sigma}](\mathbf{x}) = \nabla_x \cdot \int_{\Gamma} G(\mathbf{x}, \mathbf{y}) \vec{\sigma}(\mathbf{y}) dS(\mathbf{y}) = S^L[\nabla_{\Gamma} \cdot \vec{\sigma}](\mathbf{x}) - D^L[n \cdot \vec{\sigma}](\mathbf{x})$$

and the final formula is derived after taking the gradient of both sides.

References

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